ArogyaRoute NE

Product Requirement Document (PRD) & Technical Systems Architecture

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Target Target: Samsung Solve for Tomorrow

Version: 1.0 (Production Ready)

1. Executive Summary & Product Value (The "Why")

In the healthcare ecosystems of Northeast India—specifically spanning the complex, mountainous logistically dense corridors between Assam and Arunachal Pradesh—medical emergencies face extreme information asymmetry. Critical patients, emergency caretakers, and ambulance operators frequently travel hours to major healthcare hubs blindly, with no live assurance of resource availability. This systemic data bottleneck leads to critical losses during the "golden hour" of trauma care, severe congestion in specific municipal tertiary centers, and systemic resource exhaustion.

ArogyaRoute NE resolves this regional crisis by transforming chaos into structured coordination. It is a highly optimized, low-latency, secure full-stack web ecosystem that provides zero-friction emergency medical routing. By combining automated intelligent symptom triage, linear-regression terrain risk scoring, live verified institutional resource grids, and anti-fraud authentication layers, the platform guarantees that an emergency transit operates with maximum data clarity.

2. Product Requirements & Features

2.1 Core Feature Breakdown

- **Intelligent Visual Triage:** A natural language interface allowing users to describe localized emergency symptoms. A lightweight machine learning pipeline performs rapid processing to categorize triage severity and route patients to the correct department instantly.
- **Live Resource Grid & Directory:** An in-memory replicated ledger mapping real-time ICU beds, standard emergency beds, and active, verified on-duty specialists across private and public facilities, paired with direct operational contact pipelines.
- **Terrain & Weather Delay Mapping:** An interactive map layout rendering routes that explicitly inject variable delay scores computed from local meteorological conditions and regional landslide risk analysis.
- **Anti-Fraud Protection Layer:** A secure mobile SMS/OTP verification gate validating user authenticity prior to committing an active hospital booking request, preventing automated exhaustion of scarce clinical beds.

2.2 Constraints & Non-Functional Requirements

Metric Category	System Requirement Targets
Memory Footprint	Full system runtime must execute reliably within ≤ 90MB RAM to sustain deployment on entry-level, legacy hardware models common in rural outposts.
Processing Latency	Machine learning triage classification and route parameter evaluation must resolve under 50ms to accommodate unstable 2G/3G connectivity profiles.
Data Persistence	State data utilizes isolated thread-safe mock environments conforming completely to structured relational architectures (Users, Hospitals, Registrations), permitting a clean, 10-line drop-in configuration switch to production-grade live relational or cloud distributed engines.

3. System Architecture & Diagram (The "How")

The application architecture follows a highly modular, single-binary full-stack decoupled blueprint. It eliminates heavy runtime containerization or multi-node dependency trees to guarantee ultra-fast cold start times and negligible server resource usage.

```
[CLIENT LAYER]
  │  (Responsive HTML5 UI / Tailwind CSS / Leaflet.js Interactive Canvas)
  ▼
[SECURITY GATEWAY & INTEGRITY LAYER]
  │  (Phone Validation Engine & SMS/OTP Authenticator Protocol)
  │  (Isolation Forest Outlier Filter - Review & Traffic Protection)
  ▼
[APPLICATION CORE LAYER - PYTHON FLASK ENGINE]
  ├── /routes (HTTP Controller Enpoints: Web UI, Dynamic Bookings, Auth API)
  ├── /models (Structured Thread-Safe Collections: In-Memory Datastore)
  └── /analytics (Core Machine Learning Logic Pipelines)
        ├── Multinomial Naive Bayes Engine (Symptom Text Matrix Classification)
        └── Linear Regression Predictive Matrix (Live Weather/Terrain Delay Calibration)
  ▼
[EXTERNAL DATA CORRELATION ENGINE]
  └── OpenWeatherMap & Regional Terrestrial Landslide Risk Indices
```

Architecture Logic & Blueprint Specifications

1. **Frontend Layer:** Built with semantic HTML5, localized Tailwind utilities, and decoupled Leaflet.js mapping layers processing polyline arrays purely clientside to conserve backend execution cycles.
2. **Compute Layer:** Python Flask serving production requests wrapped under a Gunicorn WSGI container. Routes map to asynchronous, non-blocking calculations.
3. **Analytical Pipelines:** Symptoms are vectorized inline using scikit-learn CountVectorizer pipelines. Weather vectors are calculated dynamically using matrix transformations via NumPy to feed real-time route offsets.

4. Functional Validation & Release Criteria

1. **Symptom Triage Validation:** The Multinomial Naive Bayes model must maintain an F1-score of ≥ 0.85 across simulated patient input strings covering critical, urgent, and non-urgent classification boundaries.
2. **OTP Firewall Security:** Any booking intent lacking an authenticated, active session ID mapping to a verified phone signature must trigger a clean 401 Unauthorized block state.
3. **Zero Memory Overflows:** Continuous high-throughput stress tests must confirm that the total process memory space does not exceed the target threshold of 90MB RAM under concurrent evaluation loads.